

The vision: *A food-first alternative to taking digestive enzymes separately from a meal*

- The food (e.g., dressing) works as a **natural substitute** for taking something like Lactaid, Beano, Fodzyme separately from a meal. No pill required, ideally ready to eat
- The **active enzyme** is embedded directly into a food people are already eating, so support for digesting dairy, beans, and other FODMAP-heavy ingredients comes built into natural eating behavior

What enzyme formats exist today on the market?

Category	Product	Enzyme(s)	Mechanism	Enzyme Status
Pills	Lactaid	Lactase	Taken before or with dairy	Still active
Pills	Beano	Alpha-galactosidase	Taken before beans/legumes	Still active
Pills	Enzymedica (Digest Gold)	Amylase, protease, lipase, lactase, alpha-gal, cellulase + 8 more (14+ total)	Taken with each meal	Still active
Powders	Fodzyme	Fructan hydrolase, alpha-gal, lactase	Sprinkled onto food at the table	Still active
Powders	Ka'Chava	Amylase, protease, lactase, cellulase, lipase	Mixed into a meal-replacement shake	Still active
Powders	SoWell Protein	Digestive enzyme blend	Mixed into a protein shake for GLP-1 users	Still active
Food	Lactose-free dairy (Fairlife, Fage)	Lactase	<i>Added to milk before culturing — lactose is already hydrolyzed by the time it's packaged</i>	Used in food processing
Food	Naturally occurring enzymes	Bromelain, papain, diastase, microbial enzymes	<i>Native to the raw food (e.g., pineapple, papaya, honey, raw ferments); destroyed by cooking</i>	Still active
Drops	Moops	Lactase	<i>Liquid syrup added directly to food (e.g., ice cream) at the table</i>	Still active
Drops	Milkaid	Lactase	<i>Added to milk 24 hrs before drinking</i>	Used in food processing

Which food triggers are the highest priority?

Priority	Category	Common foods	Digestive limitation / substrate	Targeting enzyme
Highest Priority	FODMAPs	Milk, yogurt, ice cream	Lactose	Lactase
	FODMAPs	Beans, lentils, chickpeas	GOS	Alpha-galactosidase
Medium Priority	FODMAPs	Garlic, onion, chicory, artichoke	Inulin-type fructans	Inulinase
	Protein	Eggs, meat, dairy, legumes	Dietary protein	Protease
	Fat	Oils, butter, fatty meats	Dietary fat	Lipase
Lower Priority	FODMAPs	Wheat, barley, rye	Graminan-type fructans	Fructan hydrolase*
	FODMAPs	Apples, mangoes, honey	Excess fructose	Xylose / glucose isomerase
	FODMAPs	Mushrooms, peaches, sugar-free gum	Polyols (sorbitol / mannitol)	Under development
	Other carbs	Sugar, bread, pasta, whole grains	Sucrose, maltose, starch, pectin, insoluble fiber	Invertase, maltase, amylase, pectinase, cellulase

* Fructan hydrolase = Fodzime's enzyme blend. Source: providers.fodzime.com/research-brief

To discuss

- What is the minimum viable product to help support digestion of healthy foods (e.g., salads)?
- Are there co-formulation issues (e.g., pairing protease with lactase)?

What enzyme dosage should we target and what do products on the market use today?

Lactaid Fast Act

Lactase: 9,000 FCC / chewable tablet (2 tablets as needed)

FAST ACT
Lactase Enzyme Supplement

Supplement Facts		
Serving Size 1 Caplet	Per Caplet Amount	% Daily Value
Sodium	5 mg	<1%*
Lactase Enzyme	9000 FCC	†

*Percent Daily Values are based on a 2,000 calorie diet. †Daily Value not established.

Inactive Ingredients: Microcrystalline Cellulose, Croscarmellose Sodium, Crospovidone, Magnesium Stearate, Colloidal Silicon Dioxide



Take at First Bite | Enjoy Dairy! | Repeat

Take 1 caplet with your first bite of dairy food. If necessary, you may take 2 caplets at one time.



If you continue to eat foods containing dairy after 30–45 minutes, take another caplet.

Beano Extra Strength

Alpha-galactosidase: 800 GalU / serving

beano
EXTRA STRENGTH
Food Enzyme Dietary Supplement

Supplement Facts	
Serving Size 2 Tablets Servings Per Container 50	
Amount Per Serving	%DV
Alpha-galactosidase enzyme 800 GALU	†

†Daily Value (DV) not established

Other Ingredients: cellulose gel, mannitol, potato starch, magnesium stearate, colloidal silica, calcium gluconate.

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Enzymedica Digest Gold + ATPro

Lactase: 900 FCC | Alpha-galactosidase: 450 GalU

14+ Quality Enzymes

Supplement Facts	
SERVING SIZE: 1 CAPSULE • SERVINGS PER CONTAINER: 240	
AMOUNT PER SERVING	%DV
Digest Gold® Enzyme Blend	430 mg †
Carbohydrate-Digesting Enzymes	
Amylase Thera-Blend® (23,000 DU), Glucoamylase (50 AGU), Maltase (200 DP), Alpha Galactosidase (450 GalU), Invertase (240 SU)	
Protein-Digesting Enzyme	
Protease Thera-Blend® (80,000 HUT)	
Fiber-Digesting Enzymes	
Cellulase Thera-Blend® (3,000 CU), Beta Glucanase (25 BGU), Xylanase (550 XU), Pectinase (w/ Phytase) (45 Endo-PGU), Hemicellulase (30 HCU)	
Fat-Digesting Enzyme	
Lipase Thera-Blend® (4,000 FIP)	
Lactose-Digesting Enzyme	
Lactase (900 ALU)	
ATPro®	25 mg-300 Million LCU †
(ATP, Magnesium Citrate, Phytase, CoQ10)	

† Daily Value (DV) not established

OTHER INGREDIENTS: 100% vegetarian capsule (HPMC, water)
CONTAINS NO: Casein, crustaceans, eggs, fish, gluten, milk, nuts, sesame, shellfish, soy, wheat; Artificial colors, fillers or flavors
Recommended Use: Take 1 capsule with each meal. More if needed.
Made in USA with global ingredients

FOR CARBOHYDRATES
23K DU
AMYLASE THERA-BLEND

FOR PROTEIN
80K HUT
PROTEASE THERA-BLEND

FOR FATS
4K FIP
LIPASE THERA-BLEND

FOR DAIRY
900 ALU
LACTASE

Break Down the Entire Meal

Just 1 capsule breaks down proteins, fats, carbs, fiber & dairy

34g Fat

16g Protein

86g Carbs



ENZYMEDICA

Typical dosage ranges

Lactase: 900–18,000 FCC

Alpha-galactosidase: 450–800 GalU

To discuss: Will the taste of the food be impacted at these enzyme levels?

Is an active food-format digestive enzyme possible?

- How do we prevent enzymes from denaturing and remaining active through digestion (e.g., shelf stability, mixing with water / acid, stomach survival – very acidic environment)?
 - Encapsulation, inhibitors, stabilizing additives, separation of wet and dry ingredients until use
 - Are there certain ingredients that cannot mix with enzymes?
- What are the cost considerations (e.g., food-grade enzymes are more expensive)?
 - Where would you recommend learning more about pricing?
- Where will we source the enzymes from (encouraging enzyme science and sourcing landscape)?
- How do we ensure a great-tasting product (e.g., taste tradeoffs from fermentation)?
- How do we ensure food safety?